

SAT 10
Math Module 2

Question # ID
SAT10 M 2.1 349a5bc1

$4x + 5 = 165$
What is the solution to the given equation?

SAT10 M 2.2 a10ma202

Type of store	Average number of employees
Warehouse store	365
Department store	213
Supermarket	130

For a certain region, the table shows the average number of store employees in 2016 by type of store. Based on the table, how much greater was the average number of store employees in warehouse stores than in supermarkets?

(A) 83

(B) 152

(C) 235

(D) 495

SAT10 M 2.3 3e9eaffc

Caleb used juice to make popsicles. The function $f(x) = -5x + 30$ approximates the volume, in fluid ounces, of juice Caleb had remaining after making x popsicles. Which statement is the best interpretation of the y -intercept of the graph of $y = f(x)$ in the xy -plane in this context?

- A. Caleb used approximately 5 fluid ounces of juice for each popsicle.
- B. Caleb had approximately 5 fluid ounces of juice when he began to make the popsicles.
- C. Caleb had approximately 30 fluid ounces of juice when he began to make the popsicles.
- D. Caleb used approximately 30 fluid ounces of juice for each popsicle.

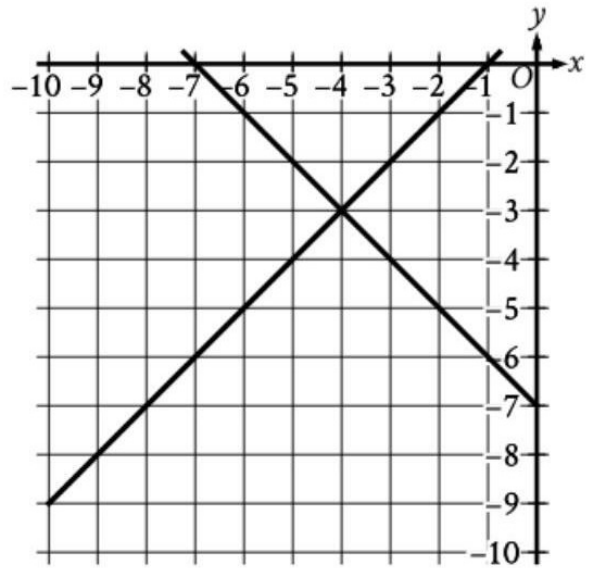
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Question # ID
SAT10 M 2.4 a10ma204

The graph of a system of linear equations is shown. What is the solution (x, y) to the system?

- (A) $(0, -7)$
- (B) $(0, -3)$
- (C) $(-4, -3)$
- (D) $(-4, 0)$



SAT10 M 2.5 a10ma205

A giant armadillo has a mass of **39** kilograms. What is the giant armadillo's mass in **grams**? (1 kilogram = 1,000 grams)

SAT10 M 2.6 Od391910

The function f is defined by $f(x) = 4x$. For what value of x does $f(x) = 8$?

SAT10 M 2.7 0bb39de4

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 18° . What is the measure of angle F ?

- A. 18°
- B. 72°
- C. 90°
- D. 162°

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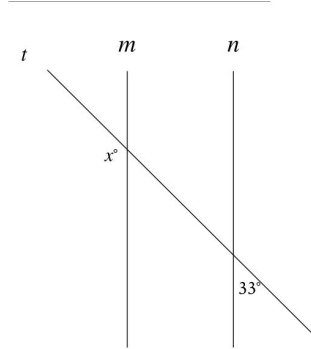
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SAT10 M 2.8

0d3f51dc



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n , and line t intersects both lines. What is the value of x ?

- A. 33
- B. 57
- C. 123
- D. 147

SAT10 M 2.9

ffb371f5

$$\begin{aligned} 3x &= 12 \\ -3x + y &= -6 \end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of y ?

- A. -3
- B. 6
- C. 18
- D. 30

SAT10 M 2.10

beb86a0c

Which expression is equivalent to $9x^2 + 5x$?

- A. $x(9x + 5)$
- B. $5x(9x + 1)$
- C. $9x(x + 5)$
- D. $x^2(9x + 5)$

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Question # ID

SAT10 M 2.11 174885f8

Jay walks at a speed of 3 miles per hour and runs at a speed of 5 miles per hour. He walks for w hours and runs for r hours for a combined total of 14 miles. Which equation represents this situation?

A. $3w + 5r = 14$

B. $\frac{1}{3}w + \frac{1}{5}r = 14$

C. $\frac{1}{3}w + \frac{1}{5}r = 112$

D. $3w + 5r = 112$

SAT10 M 2.12 02b02213

What is the perimeter, in inches, of a rectangle with a length of 4 inches and a width of 9 inches?

A. 13

B. 17

C. 22

D. 26

SAT10 M 2.13 10c448d6

A line in the xy -plane has a slope of $\frac{1}{9}$ and passes through the point $(0, 14)$. Which equation represents this line?

A. $y = -\frac{1}{9}x - 14$

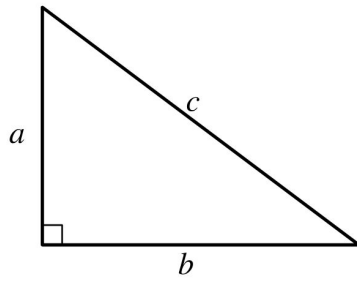
B. $y = -\frac{1}{9}x + 14$

C. $y = \frac{1}{9}x - 14$

D. $y = \frac{1}{9}x + 14$

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Question # ID
SAT10 M 2.14 c9f8d1e9



For the right triangle shown, $a = 4$ and $b = 5$. Which expression represents the value of c ?

- A. $4 + 5$
- B. $\sqrt{(4)(5)}$
- C. $\sqrt{4 + 5}$
- D. $\sqrt{4^2 + 5^2}$

SAT10 M 2.15 7038b587

Vivian bought party hats and cupcakes for \$71. Each package of party hats cost \$3, and each cupcake cost \$1. If Vivian bought 10 packages of party hats, how many cupcakes did she buy?

SAT10 M 2.16 a26c29f7

The function f is defined by $f(x) = 7x^3$. In the xy -plane, the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ down 2 units. Which equation defines function g ?

- A. $g(x) = \frac{7}{2}x^3$
- B. $g(x) = 7x^{\frac{3}{2}}$
- C. $g(x) = 7x^3 + 2$
- D. $g(x) = 7x^3 - 2$

SAT10 M 2.17 a67a439d

$$\begin{aligned}x + 7 &= 10 \\(x + 7)^2 &= y\end{aligned}$$

Which ordered pair (x, y) is a solution to the given system of equations?

- A. (3, 100)
- B. (3, 3)
- C. (3, 10)
- D. (3, 70)

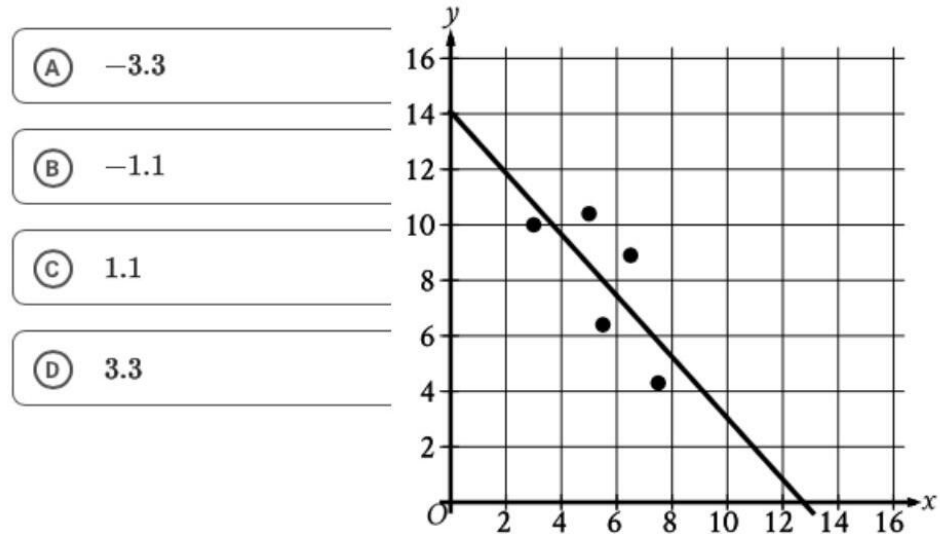
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Question # ID
SAT10 M 2.18 a10ma218

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.

Which of the following is closest to the slope of this line of best fit?



(A) -3.3

(B) -1.1

(C) 1.1

(D) 3.3

SAT10 M 2.19 5e08a055

$$y = 6x + 18$$

One of the equations in a system of two linear equations is given. The system has no solution. Which equation could be the second equation in the system?

- A. $-6x + y = 18$
- B. $-6x + y = 22$
- C. $-12x + y = 36$
- D. $-12x + y = 18$

SAT10 M 2.20 e470e19d

The function f is defined by $f(x) = 7x - 84$. What is the x -intercept of the graph of $y = f(x)$ in the xy -plane?

- A. $(-12, 0)$
- B. $(-7, 0)$
- C. $(7, 0)$
- D. $(12, 0)$

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SAT10 M 2.21	flc81b3b
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The exponential function g is defined by $g(x) = 19 \cdot a^x$, where a is a positive constant. If $g(3) = 2,375$, what is the value of $g(4)$?

SAT10 M 2.22	ba61d95f
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The population of Greenville increased by 7% from 2015 to 2016. If the 2016 population is k times the 2015 population, what is the value of k ?

- A. 0.07
- B. 0.7
- C. 1.07
- D. 1.7